

React Day 6

What are Props in React?

Props is short “properties” are the way to pass the data from one component to other component in React.

- Props are **used to send data from a parent component to a child component**.
- Props are **read-only** in the child component (they cannot be changed there).
- Props help in **making components dynamic and reusable**.

we can pass variable, object, array, func, state, children with the help of props.

How to get Input field value ?

```
{/* Input field Value */}
<input type="text" placeholder='Enter ur name' onChange={(e)=>setVal(e.target.value) } value={val} />
<button onClick={()=>setVal("")}>Clear</button>
<p>{val}</p>
```

What is a Controlled Component?

A controlled component is an input form element (like <input>, <textarea>, <select>) whose value is controlled by React state.

In controlled components:

- The value of the input is linked to a state variable.
- React controls the value, not the DOM.
- You update the value via onChange events.

Benefits

- single source of truth
- validation and manipulation before submit
- dynamic updates Values

Asan Bhasha mein

→ apne pass ek form hai jis mein ek input box hai , ab us box ka control kiske pass hai ??
React ke State ke hath mein → ye hota hai controlled Component.

```
const Controlled = () => {  
  const[name,setName] = useState('anil');  
  const[pass,setPass] = useState('');  
  const[age,setAge] = useState('');  
  
  return (  
    <div>  
      <h1>  
        Cocontrolled Component  
      </h1>  
      <form action="" method='get'>  
        <input type="text" onChange={(e)=>setName(e.target.value)} placeholder='Enter ur name' /> <br />  
        <input type="text" onChange={(e)=>setName(e.target.value)} placeholder='Enter ur age' /> <br />  
        <input type="text" onChange={(e)=>setName(e.target.value)} placeholder='Enter ur password' /> <br />  
        <button>Submit</button>  
  
        <h3>{name}</h3>  
        <h3>{age}</h3>  
        <h3>{pass}</h3>  
      </form>  
    </div>  
  )  
}
```

Handle CheckBox

```

const CheckBox = () => {

  const[skills,setSkills] = useState([]);
  const handleSkills=(event)=>{
    console.log(event.target.value,event.target.checked);
    if(event.target.checked){
      setSkills([...skills,event.target.value])
    } else{
      setSkills([...skills.filter((item)=>item !== event.target.value)])
    }
  }

  return (
    <div>

      <input type="checkbox" id='php' value="php" onChange={handleSkills}/>
      <label htmlFor="php">PHP</label>

      <input type="checkbox" id='js' value="js" onChange={handleSkills}/>
      <label htmlFor="js">JS</label>

      <input type="checkbox" id='node' value="node" onChange={handleSkills}/>
      <label htmlFor="node">node</label>

      <input type="checkbox" id='java' value="java" onChange={handleSkills}/>
      <label htmlFor="java">java</label>

      <h3>{skills.toString()}</h3>

    </div>
  )
}

```

- the handle of checkBox is done with the help of the state (here use is skills array)
- we use useState Hook here
- event.target.checked → show karnar value true ahe ki false
- event.target.value → value kay ahe te dakhavnar
- What is the **Spread Operator (...)** in JavaScript?
 - The **spread operator** is written as ... and is used to **spread out** elements of an array or object.
 - It's **not a function**, but a **feature of JavaScript** (introduced in ES6) that allows you to:
 - Copy arrays or objects
 - Merge arrays or objects
 - Pass multiple elements as function arguments
 - Update React state immutably
- How we have add element to the array
 - we have call the setSkills function and add the element
 - setSkills([...skills,event.target.value])

- How to remove element
 - we have use the filter function for that which takes the value
 - if the value is equal to the array elemnt just remove it

Radio Button

```
import React, { isValidElement, useState } from 'react'

const Radio = () => {

  const [gender, setgender] = useState("male")
  const handleGender=(event)=>{
    setgender(event.target.value)
  }

  return (
    <div>

      <h2>Selected Gender : {gender}</h2>

      <input type="radio" id='male' name='gender' onChange={handleGender} value="male"/>
      <label htmlFor="male">Male</label>

      <input type="radio" id='female' name='gender' onChange={handleGender} value="female"/>
      <label htmlFor="female">Female</label>

    </div>
  )
}
```

- name of input filled should be same because only one gender we have to choose
- if we give multiple name , it will select multiple name
- by default ke liye useState mein value daal do

DropDown

```
import React, { useState } from 'react'
|
const DropDown = () => {
  const [city, setCity] = useState('Delhi')

  return (
    <div>
      <select name="" id="" onChange={(event) => {setCity(event.target.value)}}>
        <option value="Delhi">Delhi</option>
        <option value="mumbai">mumbai</option>
        <option value="Pune">Pune</option>
      </select>
      <h2>Selected city : {city}</h2>
    </div>
  )
}

export default DropDown
```

default value ke liye state mein daal sakte hai ya fir defaultValue ye attribute hota hai ye use kr sakte hai
